

Midterm 1 SOLUTIONS, Math 170A - Section 1, Fall 2018
Instructor: Elizaveta Rebrova

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Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems. Exception: Problem 1 (see instruction in the beginning of the statement).
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. (a) (2 points) For any event A what is $\mathbf{P}(A) + \mathbf{P}(A^c)$?

Solution: Since A and A^c are always disjoint,

$$\mathbf{P}(A) + \mathbf{P}(A^c) = \mathbf{P}(A \cup A^c) = \mathbf{P}(\Omega) = 1.$$

- (b) (2 points) Let A and B be events with strictly positive probabilities. Assume that their probabilities are not equal. If $\mathbf{P}(A|B) = \mathbf{P}(B|A)$, what is the value of $\mathbf{P}(A \cap B)$?

Solution: Since A and A^c are always disjoint,

$$\mathbf{P}(A|B) = \mathbf{P}(A \cap B)/\mathbf{P}(B) = \mathbf{P}(A \cap B)/\mathbf{P}(A) = \mathbf{P}(B|A).$$

Since $\mathbf{P}(A) \neq \mathbf{P}(B)$ we have to have $\mathbf{P}(A \cap B) = 0$.

- (c) (2 points) Suppose that $B^c \subset A$, A and B are independent. What can you say about $\mathbf{P}(A)$ and $\mathbf{P}(B)$?

Solution: Since A and B are independent, then A and B^c are also independent and

$$\mathbf{P}(B^c) = \mathbf{P}(A \cap B^c) = \mathbf{P}(A) \cdot \mathbf{P}(B^c).$$

So, $(1 - \mathbf{P}(B))(1 - \mathbf{P}(A)) = 0$. Either $\mathbf{P}(B) = 1$ or $\mathbf{P}(A) = 1$.

(Other formulas with A and B obtained using both conditions (independence and inclusion) were given full credit. E.g. since $\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$ and $A \cup B = \Omega$, then $\mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A)\mathbf{P}(B) = 1$.)

- (d) (2 points) Assume that events A and B are independent. Express $\mathbf{P}(A \cup B)$ in terms of $\mathbf{P}(A)$ and $\mathbf{P}(B)$.

Solution: Since A and B are independent, then A^c and B^c are also independent. By de Morgan's law,

$$\mathbf{P}(A \cup B) = 1 - \mathbf{P}((A^c \cap B^c)^c) = 1 - \mathbf{P}(A^c) \cdot \mathbf{P}(B^c) = 1 - (1 - \mathbf{P}(A))(1 - \mathbf{P}(B)).$$

- (e) (2 points) Give an example of three sets A, B and C , so that A and B are independent, but A and B become dependent, given C . Make sure to specify the probability space (outcomes and probability law).

Solution: Many potential answers :) For example, select a number from the set $\{1, 2, \dots, 12\}$ uniformly at random. Let A be the event that the number is even, let B be the event that the number is divisible by 3, and let C be the event that the number is at most 4. Then $\mathbf{P}(A) = 1/2$, $\mathbf{P}(B) = 1/3$, $\mathbf{P}(A \cap B) = 1/6$, but $\mathbf{P}(A|C) = 1/2$, $\mathbf{P}(B|C) = 1/4$ and $\mathbf{P}(A \cap B|C) = 0$.

2. (10 points) Electric light bulbs are made at three plants:

Plant 1 (40%), Plant 2 (?? %) and Plant 3 (?? %).

The defective rates are:

Plant 1: 10%, Plant 2 (20%) and Plant 3 (5%).

Overall the defective rate is 12%. Find the proportion of light bulbs made at Plant 2.

Solution: By Total Probability theorem

$$\mathbf{P}\{\text{ lamp is defective } \} =$$

$$\mathbf{P}\{\text{ lamp is defective } | \text{ lamp is from Plant 1}\}\mathbf{P}\{\text{ lamp is from Plant 1}\} +$$

$$\mathbf{P}\{\text{ lamp is defective } | \text{ lamp is from Plant 2}\}\mathbf{P}\{\text{ lamp is from Plant 2}\} +$$

$$\mathbf{P}\{\text{ lamp is defective } | \text{ lamp is from Plant 3}\}\mathbf{P}\{\text{ lamp is from Plant 3}\}$$

So,

$$0.12 = 0.1 \cdot 0.4 + 0.2 \cdot x + 0.05 \cdot (1 - 0.4 - x).$$

Solving the equation for x , we get

$$0.2x - 0.05x = 0.12 - 0.04 - 0.05 + 0.02$$

$$0.15x = 0.05$$

$$x = 1/3 \sim 33.333\%$$

3. (10 points) There are 57 lockers in a dark hall. On a rainy day 30 students come independently and put umbrellas in these lockers randomly without looking inside (every student has one umbrella and chooses a locker uniformly at random). What is the probability that there is a locker with more than one umbrella?

Solution: Let A_i be the event that the first i students put umbrellas all in different lockers. Then $A_i = \cap_{k=1}^i \{\text{next student puts umbrella into an empty locker}\}$ and using the law of multiplication

$$\mathbf{P}(A_i) = \mathbf{P}(A_1)\mathbf{P}(A_2|A_1)\mathbf{P}(A_3|A_2) \cdot \dots \cdot \mathbf{P}(A_i|A_{i-1}).$$

Now, $\mathbf{P}(A_1) = 1$ clearly. Also $\mathbf{P}(A_2|A_1) = (57 - 1)/57$. Similarly, $\mathbf{P}(A_i|A_{i-1}) = \frac{57-i+1}{57}$.

Multiplying these all together we obtain

$$\mathbf{P}(A_{30}) = \frac{57!}{(57 - 30)!57^{30}}$$

and the probability that there are two umbrellas in the same locker is the complement of A_{30} and thus has probability

$$1 - \frac{57!}{(57 - 30)!57^{30}}.$$

4. (10 points) Let $n \geq 4$ be a positive integer. We are given a deck of n cards labeled with numbers $1, 2, \dots, n$, and we select uniformly at random two cards from this deck. Each of these two selected cards is destroyed with probability $1/2$, independently of the other card. What is the probability that neither card number 1 nor card number 2 get destroyed?

Solution: The answer clearly depends on which (if any) of cards 1 and 2 were selected initially. Let A_0 be the event that neither card 1 or 2 was selected initially; A_1 be the event that the card 1 was selected initially, but 2 wasn't, let A_2 be the event that the card 2 was selected initially but 1 wasn't, and let A_3 be the event that both cards 1 and 2 were selected initially. If B is the event that neither the card 1 nor 2 was destroyed then clearly we have

$$\mathbf{P}(B|A_0) = 1, \mathbf{P}(B|A_1) = 1/2, \mathbf{P}(B|A_2) = 1/2, \mathbf{P}(B|A_3) = 1/4.$$

Moreover, by uniform law

$$\mathbf{P}(A_0) = \frac{\binom{n-2}{2}}{\binom{n}{2}} = \frac{(n-2)(n-3)}{n(n-1)}, \mathbf{P}(A_1) = \mathbf{P}(A_2) = \frac{\binom{n-2}{1}}{\binom{n}{2}} = \frac{2(n-2)}{n(n-1)}, \mathbf{P}(A_3) = \frac{1}{\binom{n}{2}}.$$

Therefore,

$$\begin{aligned} \mathbf{P}(B) &= \mathbf{P}(B|A_0)\mathbf{P}(A_0) + \mathbf{P}(B|A_1)\mathbf{P}(A_1) + \mathbf{P}(B|A_2)\mathbf{P}(A_2) + \mathbf{P}(B|A_3)\mathbf{P}(A_3) \\ &= \frac{(n-2)(n-3)}{n(n-1)} + \frac{2(n-2)}{n(n-1)} + \frac{1}{2n(n-1)} = \frac{(n-2)(n-1) + 1/2}{n(n-1)}. \end{aligned}$$

5. (10 points) There are two stores in your neighborhood selling lottery tickets. The first store sells four types of lottery tickets priced at \$1, \$2, \$3 and \$4. The second store sells five types of lottery tickets priced at \$1, \$2, \$3, \$4 and \$5. The first store is closer to your house, so you go there two times as often (namely, the probability that you decide to go to the first store is two times probability you decide to go to the second store).

One day you had \$8 and you spent them all on 3 lottery tickets, each of different type, chosen uniformly at random in some store. What is the probability that you were in the second store?

Solution: Let A be the event that you chose first store and B the event that you chose second store, and C the event that you spent \$8 on three tickets. If you went to the first store then you could spend \$8 only if you bought 1st, 3rd and 4th ticket type, which has probability $\mathbf{P}(C|A) = 1/\binom{4}{3} = 1/4$. If you went to the second store then you could spend \$8 if you bought 1st, 3rd and 4th ticket type, or 1st, 2nd and 5th which has probability $\mathbf{P}(C|B) = 2/\binom{5}{3} = 1/5$. Also, $\mathbf{P}(A) = 2/3$ and $\mathbf{P}(B) = 1/3$ (since $\mathbf{P}(A) + \mathbf{P}(B) = 1$ and $\mathbf{P}(A) = 2\mathbf{P}(B)$). By Bayes' formula

$$\mathbf{P}(A|C) = \frac{\mathbf{P}(C|A)\mathbf{P}(A)}{\mathbf{P}(C|A)\mathbf{P}(A) + \mathbf{P}(C|B)\mathbf{P}(B)} = \frac{1/5 \cdot 1/3}{1/4 \cdot 2/3 + 1/5 \cdot 1/3} = \frac{2}{7}$$